

MAC Sub-Layer (1)

COMP90007 Internet Technologies

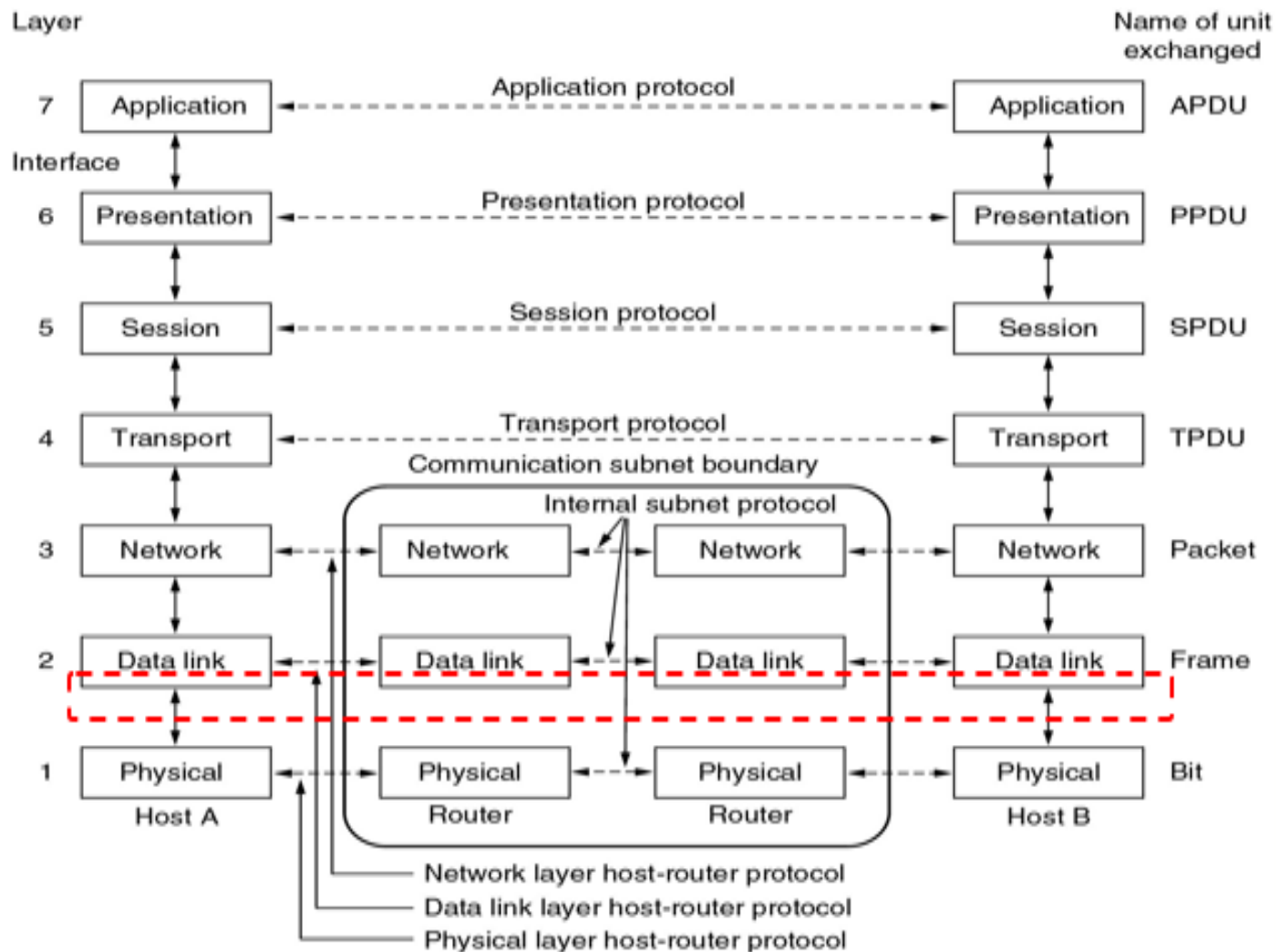
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Medium Access Control

- **Point-to-point networks** have a dedicated sender-receiver pair, which eliminates transmission contention
- **Broadcast networks** face the complex challenge of determining transmission rights
- **Medium Access Control (MAC)** sub-layer is used to assist in resolving transmission conflicts

MAC Sub-layer



Types of Channel Allocation

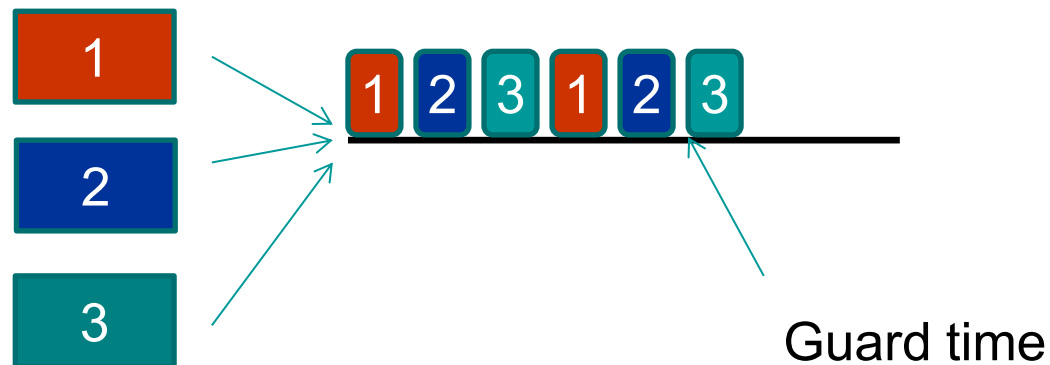
- Various methods exist for allocating a single broadcast channel amongst competing users
 - Static Channel Allocation
 - Dynamic Channel Allocation

Static Channel Allocation

- Divide a channel into segments and each user is allocated a dedicated segment for transmission
 - Time Division Multiplexing (TDM)
 - Frequency Division Multiplexing (FDM)

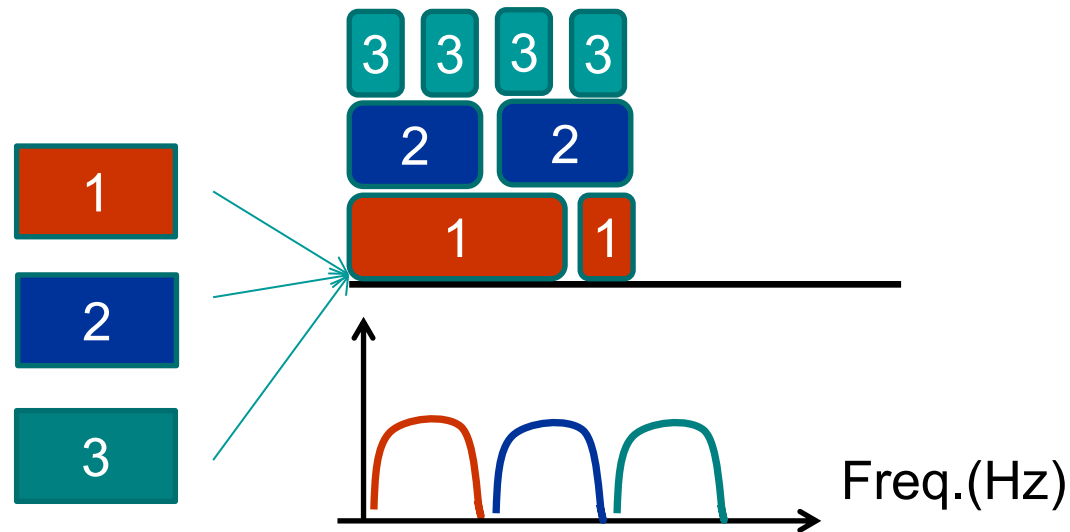
Time Division Multiplexing

- Users can send according to a fixed schedule
- Slotted access to the full speed of the channel
- e.g. 2G mobile network



Frequency Division Multiplexing

- Users can only use specific frequencies to send their data
- Continuous access with lower speed
- e.g. TV and Radio; ADSL; 4G



Static Channel Allocation

- Good for a fixed number of users, but...
- Significantly inefficient when
 - Number of senders $>$ allocated segments
 - Number of senders is not static
 - Network traffic is bursty, but static methods try to give consistent access to the network

Dynamic Channel Allocation (1)

- Channel segmentation and segment allocation are dynamic
- Assumptions for dynamic channel allocation:
 - 1) Single channel for all communication
 - 2) Independent transmission stations
 - 3) Simultaneous transmission results in collision and damaged frames

Dynamic Channel Allocation (2)

4) Time

- Continuous: Transmission can start at any time
- Slotted: Transmission can only start at predefined time intervals

5) Carrier Sense

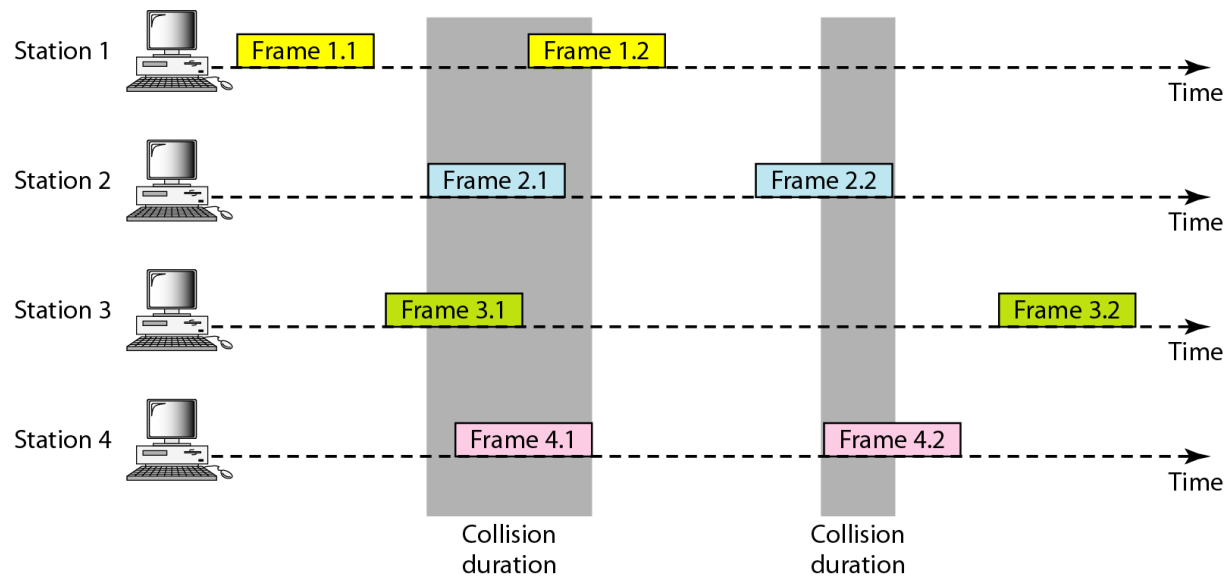
- Carrier Sense: Detecting channel activity before initiating transmission
- No Carrier Sense: Transmission occurs without detecting channel activity beforehand

Multiple Access Protocols

- Contention
 - ALOHA, Slotted ALOHA
 - Carrier Sense Multiple Access
- Collision-Free
- Limited Contention
- MACA/MACAW (for Wireless LANs)

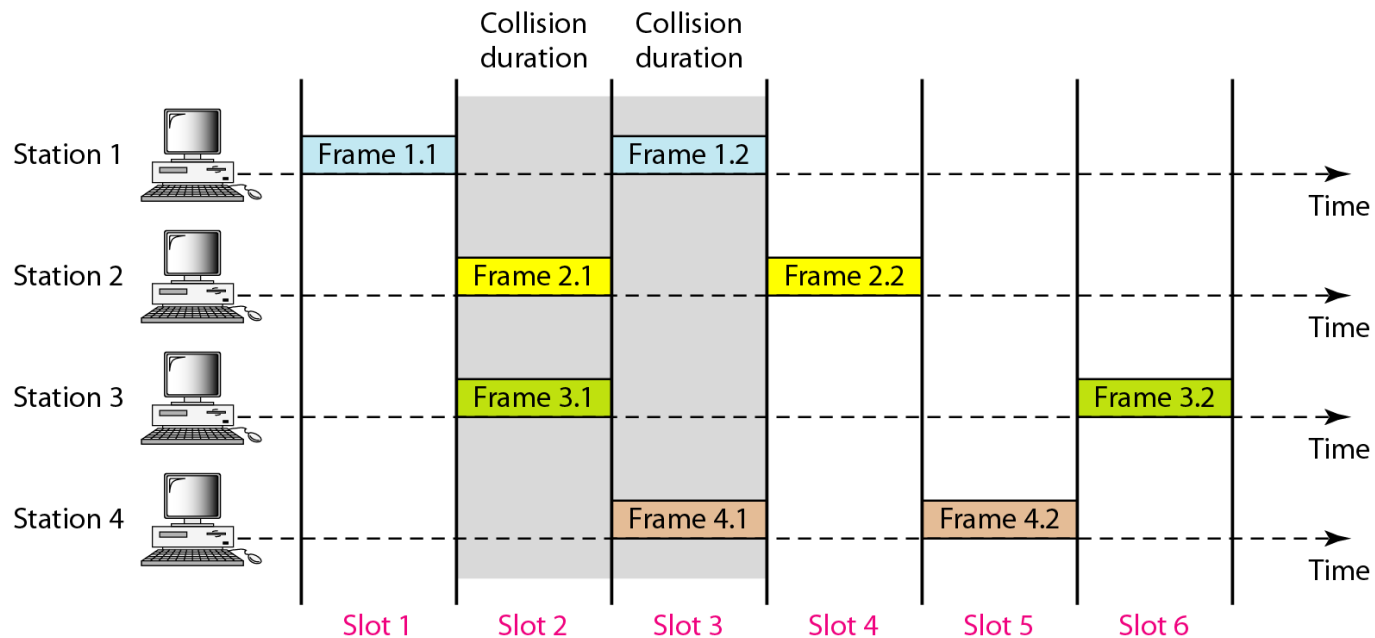
ALOHA

- Users transmit frames **whenever they have data; retry after a random time** if collisions occur or no ack is received
- Requires **no central control mechanism**
- Efficient under low load but inefficient under high traffic loads



Slotted ALOHA

- Allows users to start sending **only at the beginning of defined slots**
- Improve the efficiency of pure ALOHA by reducing the likelihood of collisions



Carrier Sense Multiple Access

- **Carrier Sense:** before transmitting, a sender first checks the channel to detect if there are any other active transmissions
- Determine transmission rights dynamically
- Protocols:
 - Persistent and Non-Persistent CSMA
 - CSMA with Collision Detection

Persistent and Non-Persistent CSMA (1)

■ 1-persistent CSMA

- Continuously sense the channel and wait until it is idle; then transmit one frame and check for collisions. If a collision occurs, wait for a random time and repeat the process.

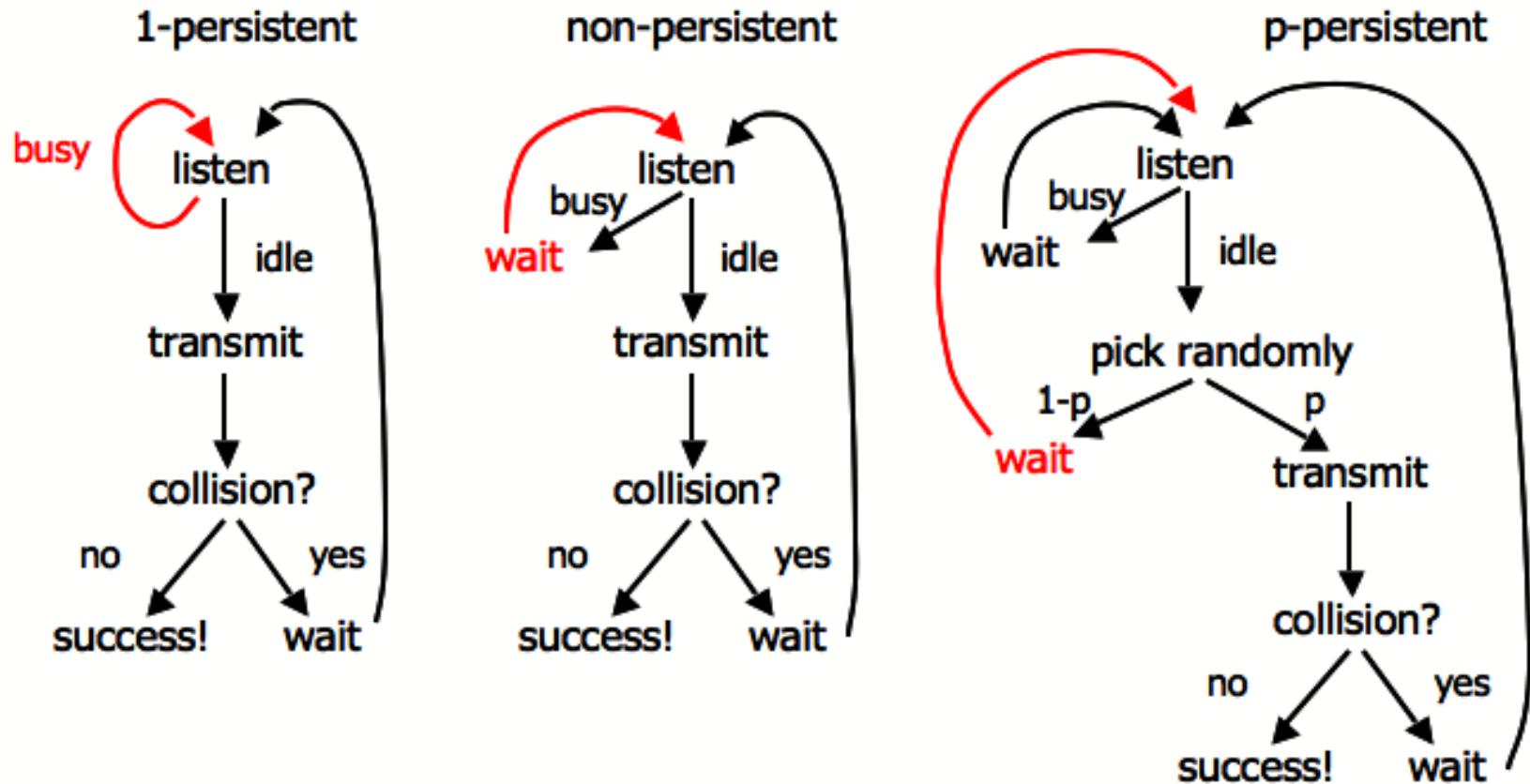
■ non-persistent CSMA

- If the channel is busy, wait a random period and check again; if idle, start transmitting. If a collision occurs, wait for a random time and repeat the process.

■ p-persistent CSMA

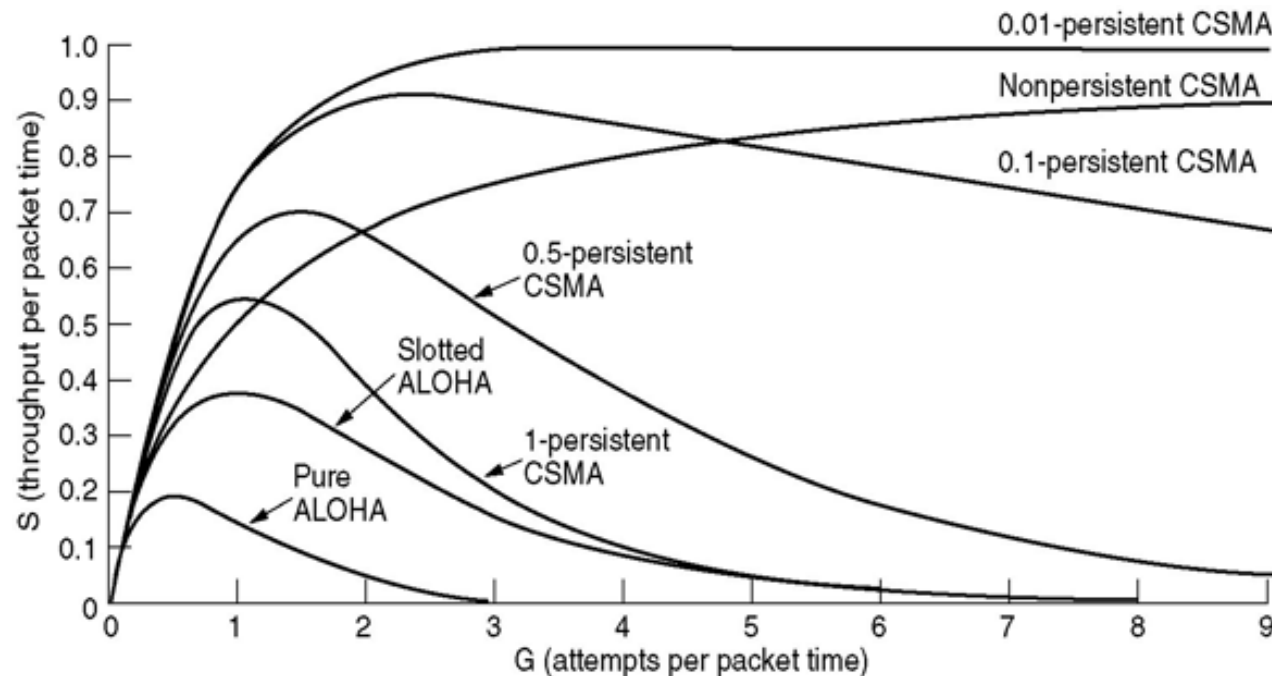
- Applies to slotted time. If the channel is idle, transmit with probability p , or defer to the next slot with probability $(1-p)$ and check again. If a collision occurs, wait for a random time and repeat the process.

Persistent and Non-Persistent CSMA (2)



CSMA Variants

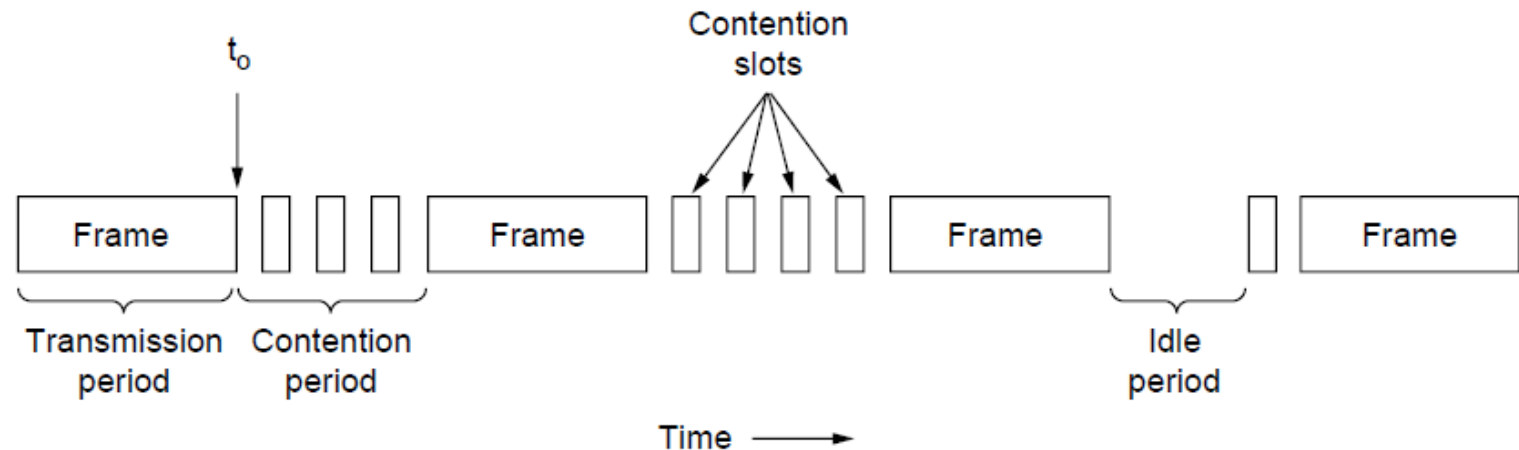
- Comparison of the efficiencies (channel utilisations)



CSMA outperforms ALOHA, and being less persistent is better under high load

CSMA with Collision Detection

- How to reduce contention times to improve performance?



- CSMA with Collision Detection
 - ❑ After collision detected, **abort transmission**, wait for a random period, and try again
 - ❑ Channel must be **continuously** monitored